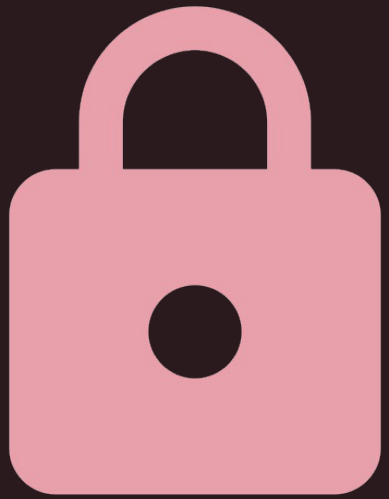
Higher National Diploma in Network Engineering, Year 2
BSc (Hons) Ethical Hacking & Network Security, Year 2

PRINCIPLES OF INFORMATION SECURITY

Secrecy with keys

Day 2 - Morning

Running case: Taxila College, a fictional college

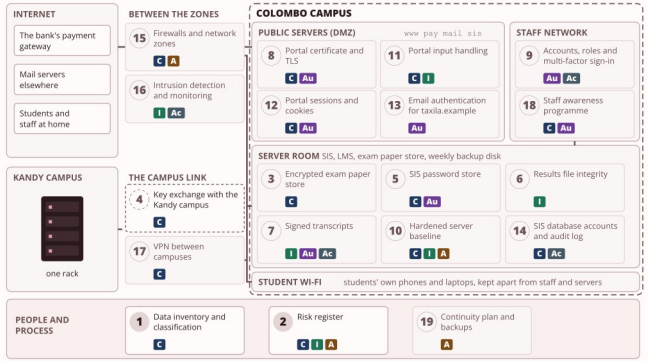


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Today

How to keep an exam paper unreadable, even if the folder leaks again.

- Plaintext, key and ciphertext - 35 min
- Why hand ciphers fail - 25 min
- Encoding, hiding and encrypting - 25 min
- Symmetric encryption and the key problem - 35 min
- Asymmetric keys and the hybrid pattern - 30 min



TODAY

4 Key exchange with the Kandy campus

Today serves one outcome

LO1 Explain the fundamentals of information security.

LO2 Discuss the key characteristics of cryptography and cryptographic systems.

LO3 Analyze the principles of encryption, decryption, hashing, and data integrity.

LO4 Outline the process of ensuring digital security.

LO5 Describe the fundamentals of threats and countermeasures.

LO6 Explain the fundamentals and process of business continuity planning.

The Registrar's new rule

RECALL

Confidentiality means only authorised people can read the data.

- Last semester a draft exam paper **leaked** from the store
- New rule: every paper unreadable without the **key**, even if the folder leaks
- The Kandy campus prints its own copies, so the key must reach Kandy

Today: how a key keeps data secret, and how to share the key.

CONCEPT 1

Plaintext, key and ciphertext

The exam paper store must stay unreadable even if everyone knows its algorithm.

TRY IT FIRST

A shift cipher and a button that tries all twenty-five keys.

35 min

Four words for secrecy

- **Plaintext**: the readable message
- **Ciphertext**: its scrambled form, which you cannot read
- **Key**: the secret that turns one into the other
- **Algorithm**: the public method, also called a **cipher**

Caesar shift, key 3: `TAXILA` becomes `WDALOD`.

Demonstration: the Caesar wheel

1. Choose a message and turn the wheel to a key
2. Read the message after the wheel
3. Press the button that tries every key
4. Watch the readable line appear among twenty-five

Twenty-five keys fall in about a millisecond.

The method can be public; the key cannot

- **Kerckhoffs's principle**: the method may be public, as long as the key stays secret
- A public method is checked by many experts; a leaked key is replaced in minutes

YOU ALREADY KNOW THIS

A combination lock's design is printed on its box. It is still safe, because only you know the combination.

MAPPING

the lock's mechanism = the algorithm, which is public · **the combination =** the key, which stays secret

× COMMON MISTAKE

Keeping the method secret is what makes it strong.

Security rests in the key; the method can be public.

CONCEPT 2

Why hand ciphers fail

A home-made cipher for passing answers would not survive one evening of counting.

TRY IT FIRST

Frequency analysis names letters in a substitution cipher.

25 min

A small key space falls to brute force

- The **key space** is the set of all possible keys
- **Brute force** tries every key, one after another
- **ROT13**, rotate by thirteen: one Caesar shift, its own inverse
- The **Vigenere** cipher shifts by a repeating keyword

Classical hand ciphers have too few keys, or leave patterns.

Demonstration: counting letters

1. Choose the longer message and read the tallest bar
2. Line the tallest bar up with e and read the result
3. Choose the short message and try again
4. Try the letter-swap cipher and fill in e

Why counting works

- **Frequency analysis** counts how often each symbol appears
- In English the commonest letter is usually **e**
- A longer message is **easier** to break: more letters to count

YOU ALREADY KNOW THIS

You keep a tally of each symbol in the message, one stroke each time you see it.

MAPPING

the tallest bar in the tally = the commonest ciphertext symbol, which is probably the letter e

T TAXILA

A home-made cipher for passing answers would not survive one evening of counting.

CONCEPT 3

Encoding, hiding and encrypting

A value written in Base64 in a configuration file protects nothing.

TRY IT FIRST

Base64, AES and image hiding side by side.

25 min

Three ways to make a message look unreadable

Encoding **No key**	Hiding **No key**	Encrypting **Key**
Base64 Changes the form of data so it travels in text. *Anyone reverses it.*	**Steganography** Hides that the message exists, for example in a picture. *Anyone who looks reads it.*	**A cipher** Scrambles the message with a key. *Only the key returns it.*

Only **encryption** needs a key.

Demonstration: encode, encrypt, hide

1. Run one line through all three
2. Get the Base64 back with no key
3. Try the encryption with no key, then with the key
4. Look at the picture, then read its hidden bits

× COMMON MISTAKE

Base64 is a kind of encryption.

Base64 has no key and anyone can reverse it.

Shorthand, a hidden note and a safe

YOU ALREADY KNOW THIS

A note can be written in shorthand, hidden inside a book, or locked in a safe.

MAPPING

shorthand = encoding, which anyone fluent reads · **the hidden note =** steganography · **the safe =** encryption, where you need the key

T TAXILA

A value written in Base64 in a configuration file protects nothing: `VGF4aWxh` is just `Taxila`.

CONCEPT 4

Symmetric encryption and the key problem

Colombo and Kandy both need the paper's key.

TRY IT FIRST

One key locks and unlocks; the wrong key fails.

35 min

One shared key for both directions

- **Symmetric encryption**: the same key locks and unlocks
- The Advanced Encryption Standard (**AES**): keys of 128, 192 or 256 bits
- The Data Encryption Standard (**DES**): 56 bits, now only history
- Each extra bit **doubles** the number of keys

Demonstration: one key, both ways

1. Make a key and encrypt a line of the paper
2. Decrypt with the same key
3. Change one character of the key and decrypt
4. Try two hundred wrong keys and read the tally

A wrong key never gives a partial answer.

How the blocks are joined

Electronic code book (ECB)

- Each 16-byte block is encrypted on its own
- Identical blocks give identical ciphertext
- The pattern of the message leaks

Cipher block chaining (CBC)

- Each block is mixed with the one before
- The first is mixed with a starting value
- Identical blocks come out different

Use CBC, not ECB: it hides the pattern.

The hard part is the key itself

- Both sides need the same secret key; sent by post, email or phone, it could be intercepted
- This is the **key distribution problem**

YOU ALREADY KNOW THIS

A locksmith cuts one key twice, so two people can open one door. You keep one copy and post the other.

MAPPING

the single cut key = the symmetric key · **posting a copy without it being intercepted =** the key distribution problem

T TAXILA

Colombo and Kandy both need the paper's key; how is it shared without interception? This is register item 4.



HEAD OF IT

Kandy prints its own copies. How do we get the key to Kandy without anyone reading it on the way?

The Head of IT, on the second day of your placement.

CONCEPT 5

Asymmetric keys and the hybrid pattern

Kandy locks the paper with Colombo's public padlock.

TRY IT FIRST

A public padlock anyone can close and a private key only you hold.

30 min

Two keys, made as a pair

- The **public key** can be given to anyone
- The **private key** is kept by its owner, never shared
- Encrypt to someone's public key; only their private key opens it

YOU ALREADY KNOW THIS

You hand out padlocks that are already open. Anyone can snap one shut. Only you have the key that opens them.

MAPPING

the open padlock = the public key, which anyone can snap shut · **the one key you keep =** the private key, which only you open with

Demonstration: the public padlock

1. Colombo makes a key pair
2. Kandy locks a short message with the public key
3. Try the public key, then the private key
4. Try to lock the whole paper, then send it the hybrid way

T TAXILA

Kandy locks the paper with Colombo's public padlock; only Colombo can open it.

Easy one way, hard the other

- Rivest, Shamir and Adleman (**RSA**): multiplying primes is easy; splitting is hard
- Toy pair: 3 x 11 = 33; public key 3, private key 7
- Encrypt 4: 4 x 4 x 4 = 64, remainder after 33 is **31**; decrypting gives 4 again
- Real keys use numbers over **six hundred digits** long
- **Diffie-Hellman** agrees a shared key in the open

The hybrid pattern

- Asymmetric methods are slow and lock only short messages
- So they set up a fast **symmetric key**, by carrying it or agreeing it
- Then the symmetric key carries the data
- Secure web traffic **agrees** its key this way, as Day 4 shows

× COMMON MISTAKE

The public key can also decrypt.

Only the matching private key decrypts what the public key locked.

The two families side by side

Symmetric

- Keys: one shared secret key
- Speed: fast
- Use: files, disks and traffic
- Weakness: the key distribution problem

Asymmetric

- Keys: a public and a private key
- Speed: much slower
- Use: carrying or agreeing a key
- Weakness: slow, short messages only

Use both: the hybrid pattern.

Common mistakes

× COMMON MISTAKE

Keeping the method secret is what makes it strong.

Security rests in the key; the method can be public.

× COMMON MISTAKE

Base64 is a kind of encryption.

Base64 has no key and anyone can reverse it.

× COMMON MISTAKE

The public key can also decrypt.

Only the matching private key decrypts what the public key locked.

Before the next day

READ	**Security in Computing, section 2.3** Then chapter 12, sections 12.2 and 12.3.
PRACTICE	**A school examinations board** Practice question 2: symmetric against asymmetric, the hybrid pattern, a stolen laptop.
PACK	**Evidence pack 2** After this afternoon's lab. *Due before the next teaching day.*

Questions & Discussion

This afternoon: ciphers by hand, then with the Advanced Encryption Standard.



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